How much solar can India's farmland host and still keep 80% of each harvest?

METHOD

Light model
shade on crops, 3 panel layouts

→

Crop model
DSSAT, 4 crops, 5 climate models

→

Yield rule
densest panels at 80% of harvest

KEY FINDINGS

HOW MUCH?

39.2 TWp

of solar fits; output: 42× India's electricity use

HOW DENSE?

52% cereals | **29%** legumes

of field under panels

Set caps by crop

BY 2050?

−1%

capacity, SSP5-8.5

Climate-robust

